A SIGNED-DIRECTION INVARIANT FOR TRIANGLE TILINGS, AND THE EXCLUSION OF PRIMES CONGRUENT TO 3 MODULO 4

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Abstract. For a triangle T and a triangle ABC, an N-*tiling* is a dissection of ABC into N triangles each congruent to T. Erdős asked which N occur; a folklore conjecture is that no prime $N \equiv 3 \pmod 4$ with $N > 3$ does (the value $N = 3$ occurs). By the classification of Laczkovich and the branch theorems of Beeson, the conjecture reduces to a tile with a $2\pi/3$ angle on a non-equilateral triangle, where a prime count forces ABC isosceles. For that isosceles case Beeson ruled out $N < 36$ and explicitly left open whether N can be prime (the smallest *known* tiling has 2673 tiles); we settle it. We introduce a translation-invariant, signed-direction tiling functional, linear in edge length and hence insensitive to non-edge-to-edge incidences, whose integrality forces, for an isosceles target, that $(c-a-b)/\sqrt{b}$ be an integer for a primitive triple with $c^2 = a^2 + ab + b^2$; an elementary lemma shows this never occurs. With a self-contained reduction of the scalene shapes, this completes the exclusion of primes $\equiv 3 \pmod 4$ exceeding 3, conditional on the cited classification of the remaining branches. Consequently the prime case of the full problem is complete: a prime p occurs if and only if $p = 2$, $p = 3$, or $p \equiv 1 \pmod 4$. Beyond primes, we determine the admissible spectrum of each sporadic $2\pi/3$ branch — for the isosceles target, writing $b = de^2$ with d squarefree, every count has the form $N = dw^2(a + 2b)$ with $e \mid w(c - a - b)$ — bounding the realizable set from outside; and a lattice theorem determines the refined spectrum exactly, exhibiting admissible values beyond Zhang's constructed families (the smallest being $N = 354$). Combining the spectra with the published branch theorems and an exhaustive exact search of the surviving finite instances, we settle every previously undetermined value up to 41: no triangle can be cut into 14, 15, 21, 22, 30, 33, 35, 38, or 39 congruent triangles; $N = 46$, previously the sharpest test of Zhang's completeness conjecture, is likewise excluded; and membership in the tile-count set is shown to be decidable. The arithmetic and combinatorial layer is machine-checked in Lean 4 with Mathlib.

1. Introduction

Let T be a triangle (the *tile*). An N-*tiling* of a triangle ABC is a way of writing ABC as a union of N triangles, each congruent to T, with pairwise disjoint interiors. We do *not* assume the tiling is edge-to-edge: a vertex of one tile may lie in the relative interior of an edge of another. Erdős asked (reported by Soifer [13]) to determine the set of N for which some triangle admits an N-tiling by some tile. Every perfect square occurs (any triangle, by the n^2 subdivision), and Soifer observed that $n^2 + m^2$, $2n^2$, $3n^2$ and $6n^2$ occur as well; in particular $3 = 3 \cdot 1^2$ occurs. The outstanding question is whether the primes $\equiv 3 \pmod 4$ that exceed 3, which are exactly the odd primes that are neither sums of two squares nor of the form $3n^2$, are excluded.

Theorem 1. *No prime $N \equiv 3 \pmod 4$ with $N > 3$ is a number of congruent triangles into which some triangle can be cut. In particular, no triangle can be cut into 19 congruent triangles.*

The restriction $N > 3$ is necessary: an equilateral triangle joined to its centre splits into 3 congruent triangles, so $N = 3$ occurs.

The proof rests on Laczkovich's classification of triangle tilings [9, 10], organized by Beeson across a series of papers [1, 2, 3, 4, 5]. For a prime number of tiles every branch of the classification

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is excluded by an existing theorem, with one exception: a tile having a $2\pi/3$ angle, tiling a non-equilateral triangle. We treat that branch.

This branch is the active frontier. Zhang [7] classifies the non-equilateral $2\pi/3$ shapes into six sporadic similarity types (Section 2 below), constructs families of tilings for each, and conjectures that the constructed N, all composite, are the only ones; the present results prove the no-prime half of that conjecture for the isosceles type, and reduce the scalene types directly. For the isosceles type Beeson [3] proved that no tiling exists with $N < 36$ and *explicitly left open whether N can be prime*; the smallest tiling presently known has 2673 tiles (Herdt, reported in [3]). The invariant below settles the question, excluding *every* prime. The adjacent Erdős Problem 633, on tilings into a square number of triangles, was recently settled by Beeson, Laczkovich and Zhang [8].

Two reductions prepare it. First, by the rationality theorem of Beeson and Zhang [6], such a tile has commensurable sides, so after scaling we may take it integer-sided with $c^2 = a^2+ab+b^2$. Second, an elementary area-and-incidence argument shows that a *prime* number of such tiles forces ABC to be *isosceles* (Section 3).

The heart of the paper is Section 4. The natural parity argument for a $2\pi/3$ tile (a black/white colouring of the tiles) fails, because a vertex of the tiling may be surrounded by an even number of tiles, so no consistent two-colouring exists. We replace it by a functional Φ that assigns to a directed edge of direction $\theta = j\frac{\pi}{3} + k\alpha$ the weight (length) $\cdot\,(-1)^j$. Two features make it work: $\Phi(\theta+\pi) = -\Phi(\theta)$, so interior edges cancel; and the weight is *linear in edge length*, so the cancellation is unaffected by non-edge-to-edge incidences, where one long edge meets several shorter ones summing to it. Summing over the tiles, Φ of the boundary equals an integer multiple of a fixed tile-value $c+a-b$. For an isosceles target this forces $(c-a-b)/\sqrt{b} \in \mathbb{Z}$, which never holds (Section 5). Assembling the branches proves Theorem 1 (Section 6).

The combinatorial claims are verified by the scripts in the accompanying repository, and the entire arithmetic layer, including Lemma 6 and the reduction of Section 3, is formalized and machine-checked in Lean 4 with Mathlib; see Section 8.

2. Preliminaries

The classification. Fix a prime $N = p > 3$ with $p \equiv 3 \pmod 4$. By Laczkovich [9, 10], every N-tiling of a triangle falls into one of finitely many types of pairs (ABC, T). The relevant dichotomy, in Beeson's organization [4], is: either the angles of T are pairwise commensurable with π, or they are not. If they are commensurable, then N is a square, a sum of two squares, or 2, 3 or 6 times a square [4, Thm. 3 and Table 3]; but $p > 3$ with $p \equiv 3 \pmod 4$ is a square only if p is one (it is not), a sum of two squares only if $p \not\equiv 3 \pmod 4$, and $2n^2, 3n^2, 6n^2$ only if p is even or $3 \mid p$ (here p is odd and $p > 3$), so p is none of these (it is precisely for $p = 3 = 3\cdot 1^2$ that this step would fail). If the angles are not commensurable, then, after the right-angle and similar-tile cases, which give N a sum of two squares or $N = n^2$ [1, 12], the tile has a special angle and falls into one of: $3\alpha + 2\beta = \pi$, for which N is not prime [2]; an isosceles tile with $\gamma = 2\alpha$, for which N is not squarefree [3, Thm. 11.7]; or the tile has an angle equal to $\pi/3$ or $2\pi/3$. The case of a $\pi/3$ or $2\pi/3$ tile on an *equilateral* triangle gives N not prime [5]. What remains for a prime N is a tile with a $\pi/3$ or $2\pi/3$ angle on a *non-equilateral* triangle. The vertex enumeration of Section 2 shows the $\pi/3$ case admits only equilateral or tile-similar ABC, already covered; so the sole remaining branch is a $2\pi/3$ tile, which we now fix.

The tile. Throughout, T has angles (α, β, γ) opposite sides (a, b, c), with $\gamma = 2\pi/3$, hence $\alpha + \beta = \pi/3$; we treat the incommensurable case, α an irrational multiple of π. By Beeson–Zhang [6], T has commensurable sides, so we scale to

$$(a,b,c) \in \mathbb{Z}^3, \qquad \gcd(a,b,c) = 1, \qquad c^2 = a^2+ab+b^2 \tag{1}$$

(the law of cosines at $\gamma = 2\pi/3$). We call such a triple a *primitive* $120°$*-triple*. If a prime p divides two of a, b, c it divides the third (by (1)), so (1) gives $\gcd(a, b) = \gcd(a, c) = \gcd(b, c) = 1$. One computes

$$\sin\alpha = \tfrac{a\sqrt{3}}{2c}, \quad \cos\alpha = \tfrac{2b+a}{2c}, \qquad \sin\beta = \tfrac{b\sqrt{3}}{2c}, \quad \cos\beta = \tfrac{2a+b}{2c}. \tag{2}$$

The direction grid. Set $G = \{\, j\frac{\pi}{3} + k\alpha : j, k \in \mathbb{Z} \,\} \pmod{2\pi}$. We claim every edge of every tile in a tiling points in a direction lying in G, for a single fixed reference. Fix the reference along an edge of one tile; its three edge-directions are then in G, since the tile angles $\alpha, \beta = \frac{\pi}{3} - \alpha, \gamma = \frac{2\pi}{3}$ all lie in $\mathbb{Z}\alpha + \mathbb{Z}\frac{\pi}{3}$. Now argue by adjacency. The tiling fills a connected region, so any tile t' shares at least a boundary point P with the union of the tiles already placed; at P the surrounding corner angles each belong to $\{\alpha, \beta, \gamma\}$, and a directed edge of t' at P differs from a known grid direction at P by a sum of such angles, hence lies in G; the remaining edges of t' then differ from it by $\mathbb{Z}$-combinations of $\alpha, \frac{\pi}{3}$, so all of t' is in G. This uses only the angles meeting at a point and so applies verbatim at non-edge-to-edge incidences, where P lies in the relative interior of an edge (the angles there sum to π rather than 2π). Since α/π is irrational, $\frac{\pi}{3}$ and α are $\mathbb{Q}$-linearly independent, so the pair (j, k) is determined by the direction (and $(-1)^j$ by the direction alone, as $2\pi = 6 \cdot \frac{\pi}{3}$).

Vertex types and shapes of ABC. A corner of ABC is filled by tile-corners: if $P, Q, R \geq 0$ (not all 0) of them are α-, β-, γ-corners, the corner measures $P\alpha + Q\beta + R\gamma = (P - Q)\alpha + (Q + 2R)\frac{\pi}{3}$. Write $(m, k) := (P - Q,\ Q + 2R)$. Since $Q \equiv k \pmod 2$, $0 \leq Q \leq k$, and $P = m + Q \geq 0$, the pair (m, k) is *realizable* exactly when

$$k \geq 1 \text{ and } m \geq -k, \qquad \text{or} \qquad k = 0 \text{ and } m \geq 1. \tag{3}$$

The three corners of ABC have angles $m_i\alpha + k_i\frac{\pi}{3}$ summing to $\pi = 3 \cdot \frac{\pi}{3}$; as α is irrational this forces

$$\textstyle\sum_i m_i = 0, \qquad \sum_i k_i = 3. \tag{4}$$

By (3) each $m_i + k_i \geq 0$, and by (4) $\sum_i (m_i + k_i) = 3$; hence each $m_i + k_i \in \{0, 1, 2, 3\}$ and $m_i \in [-k_i,\ 3 - k_i] \subseteq [-3, 3]$. The enumeration is therefore finite and *independent of* α. Checking the finitely many triples $\{(m_i, k_i)\}$ satisfying (3), (4) and $0 < m_i\alpha + k_i\frac{\pi}{3} < \pi$ (done in exact arithmetic in Section 8) leaves exactly eleven shapes of ABC: the equilateral triangle; two isosceles triangles, with base angles α (apex $\alpha + 3\beta$) and β (apex $\beta + 3\alpha$); and eight scalene triangles, namely

$$F_1 = (\alpha, \alpha + \beta, \alpha + 2\beta), \quad F_2 = (2\alpha, 2\beta, \alpha + \beta), \quad F_3 = (\alpha, 2\alpha, 3\beta), \quad F_4 = (\alpha, 2\beta, 2\alpha + \beta),$$

their reflections under $\alpha \leftrightarrow \beta$ (with F_2 self-paired), and the tile-similar triangle $(\alpha, \beta, 2\pi/3)$. Since α/π is irrational, distinct pairs (m, k) give distinct angles, so the labels *equilateral*, *isosceles*, *scalene* attached to these types are unambiguous: no accidental coincidence of angles can occur, as any such coincidence would force $\alpha \in \mathbb{Q}\pi$. These types coincide with the six sporadic non-equilateral $2\pi/3$ similarity types of Zhang [7].

The area identity. If ABC is similar to a fixed shape whose primitive integer side-proportion vector is D (gcd of its entries 1), and ABC is N-tiled, then each side of ABC is a disjoint union of whole tile-edges (a tile meeting a straight portion of the boundary has a whole side on it), hence an integer; so the sides of ABC are $k \cdot D$ for an integer $k \geq 1$, and

$$N = \frac{\text{Area}(ABC)}{\text{Area}(T)} = k^2 N_0, \qquad N_0 := \frac{\text{Area}(D)}{\text{Area}(T)}, \qquad \text{Area}(T) = \tfrac{\sqrt{3}}{4}ab. \tag{5}$$

This is a necessary condition on N.

3. Reduction to the isosceles case

Proposition 2. *Let T be a tile with angles $(\alpha, \beta, 2\pi/3)$, α/π irrational. If T N-tiles a triangle and N is prime, then ABC is isosceles and not equilateral.*

Proof. By Section 2, the corner-type triple of ABC is the equilateral one, the tile-similar one, one of $F_1, \dots, F_4$ or their reflections, or one of the two isosceles ones. The equilateral case gives N not prime [5, Thm. 6]; the tile-similar case gives $N = n^2$ [1, 12], not prime. For the remaining non-isosceles shapes we compute N_0 from (5); in each case $N = k^2 N_0$ is composite.

F_2 and F_4. The side-proportion vector of F_2 is $\big(a(a + 2b),\, b(2a + b),\, c^2\big)$ with the $\pi/3$ vertex between the first two sides, giving $N_0 = (a + 2b)(2a + b)$. For F_4 one checks the side-vector $\big(ac,\, b(2a + b),\, (a + b)c\big)$ has gcd 1 (any common prime g divides c and then $g^2 \mid 3b^2$, forcing $g = 1$), giving $N_0 = (a + b)(2a + b)$. Both are products of two integer factors each at least 3, hence composite, so $N = k^2 N_0$ is composite.

F_3. The law of sines gives the side-proportion vector $\big(ac^2,\, a(a+2b)c,\, 3ab(a+b)\big)$, whose entries share the factor a; dividing it out leaves $\big(c^2,\, (a + 2b)c,\, 3b(a + b)\big)$, and $N_0 = 3(a + b)(a + 2b)$. This is divisible by 3, hence composite, so $N \equiv 0 \pmod 3$ is composite.

F_1. The side-vector is $(a, c, a + b)$ with the $\pi/3$ vertex between the sides a and $a + b$, so $N_0 = (a+b)/b$ and $N = k^2(a+b)/b$. As $\gcd(b, a+b) = 1$, integrality forces $b \mid k^2$, so $N = t(a+b)$ with $t = k^2/b \in \mathbb{Z}_{\geq 1}$. It remains to show $a + b$ is composite. Suppose $a + b = p$ is prime. Since $(a + b)^2 - c^2 = ab$ by (1), we have $ab = p^2 - c^2$, so a, b are the roots of $x^2 - px + (p^2 - c^2)$, whose discriminant $4c^2 - 3p^2$ must be a perfect square d^2; then $3p^2 = (2c - d)(2c + d)$. With $2c - d < 2c + d$ the factorizations of $3p^2$ into two positive factors are $(1, 3p^2)$, $(3, p^2)$, $(p, 3p)$, and in each the smaller root $\frac{p-d}{2}$ is not a positive integer: in the first, $d = (3p^2 - 1)/2 > p$, so the root is negative; in the second, $d = (p^2 - 3)/2$ and the root $\frac{p-d}{2} = -\frac{(p-3)(p+1)}{4}$ is negative for $p > 3$; in the third, $d = p$ gives $c = p$ and $ab = p^2 - c^2 = 0$. (The case $p \leq 3$ is impossible since $a + b \geq 8$ for a primitive 120°-triple.) Hence $a + b \geq 8$ is composite, and $N = t(a + b)$ is composite.

The reflected shapes are handled by $\alpha \leftrightarrow \beta$ (which swaps $a \leftrightarrow b$ and leaves each N_0 above of the same form). So a prime N forces the corner-type triple of ABC to be one of the two isosceles types; those have angles $(\varphi, \varphi, \pi - 2\varphi)$ with $\varphi \in \{\alpha, \beta\}$, and they are never equilateral ($\varphi = \frac{\pi}{3}$ would force $\beta = 0$ or $\alpha = 0$). Hence ABC is isosceles and not equilateral. $\square$

4. The signed-direction invariant

Fix the reference so the direction grid is $G = \{\, j\frac{\pi}{3} + k\alpha \,\}$. Because $\beta = \frac{\pi}{3} - \alpha$, every grid direction also has the form $(j+k)\frac{\pi}{3} - k\beta$, so we may read off the coefficient of $\frac{\pi}{3}$ in two coordinate systems. Define, for a direction $\theta \in G$,

$$f_\alpha(\theta) = (-1)^j \quad (\theta = j\tfrac{\pi}{3} + k\alpha), \qquad f_\beta(\theta) = (-1)^{j'} \quad (\theta = j'\tfrac{\pi}{3} + k'\beta).$$

Both are well defined (Section 2), and both satisfy

$$f(\theta + \pi) = -f(\theta), \tag{6}$$

since adding $\pi = 3 \cdot \frac{\pi}{3}$ increases the $\frac{\pi}{3}$-coefficient by 3. They are genuinely different functions: $f_\beta = (-1)^k f_\alpha$.

For $f \in \{f_\alpha, f_\beta\}$ and a directed edge of length L and direction θ, assign the weight $L\, f(\theta)$. For a tile t with boundary traversed counterclockwise, put $C_f(t) = \sum_{\text{edges}} L\, f(\theta)$, and for the boundary of ABC put $\Phi_f(\partial ABC) = \sum_{\text{sides}} L\, f(\theta)$ (counterclockwise).

Lemma 3 (Cancellation)**.** *For each $f \in \{f_\alpha, f_\beta\}$, $\displaystyle\sum_{\textit{tiles}} C_f(t) = \Phi_f(\partial ABC)$.*

Proof. For a grid direction θ let $\Lambda(\theta)$ be the total length of all directed tile-edges of direction θ, summed over every tile (counterclockwise) and every edge; then, grouping the double sum $\sum_t C_f(t)$ by direction,

$$\sum_{\text{tiles}} C_f(t) = \sum_{\theta} \Lambda(\theta)\, f(\theta). \tag{7}$$

Split each tile-edge into its maximal sub-segments that are either interior to ABC or contained in ∂ABC; this only refines the sums and changes nothing. Consider the finite set of maximal straight segments S carved out on the tiling's edges. If S lies in the interior of ABC, a neighbourhood of a generic point of S is a disc split by S into two half-discs, each tiled; the tiles on one side meet S in directed edges of total length $|S|$ pointing one way, say θ, and those on the other side in directed edges of total length $|S|$ pointing $\theta+\pi$ (the two subdivisions may differ, a non-edge-to-edge incidence, but each side covers S exactly once because the tiles cover ABC with disjoint interiors). If instead $S \subseteq \partial ABC$, only the single tile inside ABC meets it, contributing length $|S|$ in the counterclockwise direction of ∂ABC. Writing $\Lambda = \Lambda_{\text{int}} + \Lambda_{\text{bd}}$ accordingly, interior segments give $\Lambda_{\text{int}}(\theta) = \Lambda_{\text{int}}(\theta+\pi)$ for every θ, while $\Lambda_{\text{bd}}(\theta)$ is the counterclockwise boundary length in direction θ. Substituting into (7) and pairing θ with $\theta+\pi$,

$$\sum_{\{\theta,\theta+\pi\}} \Lambda_{\text{int}}(\theta)\big(f(\theta)+f(\theta+\pi)\big) = 0$$

by (6), so $\sum_{\text{tiles}} C_f(t) = \sum_\theta \Lambda_{\text{bd}}(\theta) f(\theta) = \Phi_f(\partial ABC)$. □

Lemma 4 (Tile value)**.** *For every placement of T, $C_{f_\alpha}(t) = \pm(c+a-b)$ and $C_{f_\beta}(t) = \pm(c+b-a)$.*

Proof. Take a positively oriented placement and label the vertices V_1, V_2, V_3 in counterclockwise order with interior angles α, β, γ respectively; let e_i be the directed edge from V_i to V_{i+1} (indices mod 3). Then e_1, e_2, e_3 are opposite V_3, V_1, V_2, so they have lengths c, a, b. Traversing counterclockwise turns left at V_{i+1} by the exterior angle $\pi-$(angle at V_{i+1}), so $\operatorname{dir}(e_2) = \operatorname{dir}(e_1) + (\pi-\beta)$ and $\operatorname{dir}(e_3) = \operatorname{dir}(e_2) + (\pi-\gamma)$. Now π adds 3 to the $\frac{\pi}{3}$-coefficient j, while α adds 0, $\beta = \frac{\pi}{3} - \alpha$ adds 1, and $\gamma = \frac{2\pi}{3}$ adds 2; so $\pi - \beta$ adds $3-1=2$ and $\pi-\gamma$ adds $3-2=1$. Writing j_0 for the $\frac{\pi}{3}$-coefficient of $\operatorname{dir}(e_1)$, the three coefficients are j_0, j_0+2, j_0+3, so $f_\alpha = (-1)^{j_0}$ on e_1, e_2 and $-(-1)^{j_0}$ on e_3. As e_1, e_2, e_3 carry lengths c, a, b,

$$C_{f_\alpha}(t) = (-1)^{j_0}\big(c+a-b\big).$$

Any placement is obtained from this one by a grid rotation $m\alpha + n\frac{\pi}{3}$, which shifts every j by n and multiplies C_{f_α} by $(-1)^n$, possibly followed by a reflection, which reverses the traversal and negates C_{f_α}; hence $C_{f_\alpha}(t) = \pm(c+a-b)$ in every case. Repeating in the β-frame (where $\gamma = \frac{2\pi}{3}$ adds 0 to j' and α adds 1) makes the a-edge the odd one out, giving $C_{f_\beta}(t) = \pm(c+b-a)$. □

Corollary 5 (Integrality and parity)**.** *Each of $M_\alpha = \dfrac{\Phi_{f_\alpha}(\partial ABC)}{c+a-b}$ and $M_\beta = \dfrac{\Phi_{f_\beta}(\partial ABC)}{c+b-a}$ is an integer, and moreover*

$$M_\alpha \equiv M_\beta \equiv N \pmod 2.$$

They are separate necessary conditions: each follows from Lemma 3 for its own f, so a single failure forbids a tiling.

Proof. By Lemmas 3 and 4, $\Phi_{f_\alpha}(\partial ABC) = \sum_t C_{f_\alpha}(t) = (c+a-b)\sum_t \varepsilon_t$ with $\varepsilon_t \in \{\pm 1\}$, so $M_\alpha = \sum_t \varepsilon_t$ is a sum of N signs, whence $M_\alpha \equiv N \pmod 2$; likewise for f_β. □

5. The isosceles obstruction

Lemma 6 (Non-integrality)**.** *Let a, k be positive integers with $\gcd(a, k) = 1$, put $b = k^2$, and let $c > 0$ satisfy $c^2 = a^2 + ab + b^2$. Then $k \nmid (a + b - c)$.*

Proof. From $c^2 = a^2 + ab + b^2$ we get $c > a$ and, since $c^2 < (a + b)^2$, also $c < a + b$. Suppose $k \mid (a + b - c)$. As $b = k^2 \equiv 0 \pmod k$, this gives $c \equiv a \pmod k$; with $0 < c - a < b = k^2$ we may write $c = a + kt$, $1 \le t \le k - 1$. Substituting into $c^2 = a^2 + ak^2 + k^4$ and dividing by k,

$$2at + kt^2 = ak + k^3, \qquad \text{that is} \qquad a(2t - k) = k(k - t)(k + t).$$

Then $k \mid a(2t - k)$, and $\gcd(a, k) = 1$ gives $k \mid (2t - k)$, hence $k \mid 2t$. But $2 \le 2t \le 2k - 2$ contains no multiple of k other than k itself, so $2t = k$. The left side is then $a(2t-k) = 0$, while the right side is $k \cdot \frac{k}{2} \cdot \frac{3k}{2} = \frac{3k^3}{4} > 0$, a contradiction. Hence $k \nmid (a + b - c)$. □

Theorem 7. *Let T be a tile with angles $(\alpha, \beta, 2\pi/3)$, α/π irrational. Then no prime number of copies of T tiles an isosceles, non-equilateral triangle.*

Remark 8. The irrationality hypothesis cannot be dropped: the commensurable tile $(\frac{\pi}{6}, \frac{\pi}{6}, \frac{2\pi}{3})$ tiles an equilateral (hence isosceles) triangle with $N = 3$, by the fan from the centre; that tile lives in the commensurable branch of the classification, where $N = 3n^2$ is allowed. The equilateral target for an *incommensurable* $2\pi/3$ tile is excluded for primes by Beeson [5], and is covered in the assembly of Section 6; it is not part of this theorem.

Proof. ABC has exactly two equal corner angles. Each corner angle is $m\alpha + k\frac{\pi}{3}$ (Section 2); since α/π is irrational, two corner angles are equal only if their pairs (m, k) coincide. So the corner-type triple of ABC has exactly one repeated type, and among the eleven triples of Section 2 exactly the two isosceles ones qualify. Hence the base angle is α or β. We use the invariant generated by the base angle, so that the equal sides land on $\frac{\pi}{3}$-coefficient 3 (odd) and pick up $f = -1$.

If the base angle is α, then $ABC = k(c, c, 2b + a)$ (equal sides kc, base $k(2b + a)$): the base is at direction 0 ($f_\alpha = +1$) and the equal sides at $\pi - \alpha = 3\frac{\pi}{3} - \alpha$ ($f_\alpha = -1$), so $\Phi_{f_\alpha}(\partial ABC) = k(2b+a) - 2kc = k(2b+a-2c)$. The tiling count is $N = k^2(a+2b)/b$; since N is prime and $\gcd(a+2b, b) = 1$, we have $k^2 = b$, i.e. $k = \sqrt{b}$. By Corollary 5 and $c^2 = a^2+ab+b^2$,

$$\frac{k(2b + a - 2c)}{c + a - b} = \frac{c - a - b}{k} \in \mathbb{Z}, \qquad k = \sqrt{b},$$

so $k \mid (a+b-c)$, contradicting Lemma 6. If the base angle is β, then $ABC = k(c, c, 2a + b)$, the equal sides are at $\pi - \beta$, and using f_β (for which the equal sides have $\frac{\pi}{3}$-coefficient 3) the same computation gives, with $k = \sqrt{a}$, $(c - a - b)/k \in \mathbb{Z}$, again contradicting Lemma 6 (the integer $a + b - c$ is symmetric in a, b). In both cases no tiling exists. □

Remark 9. The role of the two invariants is essential. For the base-angle-β shape, f_α places the equal sides at $\pi - \beta = \frac{2\pi}{3} + \alpha$, an *even* $\frac{\pi}{3}$-coefficient, so f_α gives an integer value and does not obstruct; f_β does. Each invariant is a necessary condition in its own right (Corollary 5), so using the one that fails is legitimate, not a choice of convenient frame: on any genuine tiling both invariants return an integer.

6. Resolution of the problem

Proof of Theorem 1. Let $p > 3$ be prime with $p \equiv 3 \pmod 4$, and suppose some triangle is p-tiled. By the classification (Section 2) the tile is in one of the listed types, and every type other than a $2\pi/3$ tile on a non-equilateral triangle excludes such a p by an existing theorem [12, 1, 4, 2, 3, 5]. For a $2\pi/3$ tile, Proposition 2 forces ABC isosceles and not equilateral, and

Theorem 7 excludes that. Hence no such tiling exists. Finally $19 > 3$ is prime and $\equiv 3 \pmod 4$, so no triangle is cut into 19 congruent triangles. $\square$

7. Towards the full problem

Erdős's question asks for *all* N. Theorem 1 settles the negative side for primes; this section records what the present methods contribute to the full determination: a complete answer for primes, and, for each sporadic $2\pi/3$ branch, a determination of its *admissible spectrum* — the set of counts compatible with the area and invariant conditions.

Theorem 10 (The prime case, complete)**.** *A prime p is a number of congruent triangles into which some triangle can be cut if and only if $p = 2$, $p = 3$, or $p \equiv 1 \pmod 4$.*

Proof. If. For $p = 2$, the altitude from the right angle splits an isosceles right triangle into two congruent triangles. For $p = 3$, join the centre of an equilateral triangle to its vertices. For $p \equiv 1 \pmod 4$, Fermat's two-squares theorem gives $p = e^2 + f^2$ with $e, f \geq 1$; splitting a right triangle with legs e, f by the altitude from the right angle and tiling the two similar pieces quadratically with e^2 and f^2 congruent copies realizes $N = e^2 + f^2$ (the biquadratic tiling [13, 3]). *Only if.* Theorem 1, whose dependencies this direction inherits. $\square$

Theorem 11 (Isosceles admissible spectrum)**.** *Let (a, b, c) be a primitive $120°$-triple and write $b = de^2$ with d squarefree. Every N-tiling of the base-α isosceles target has scale $k = dew$ for a positive integer w, and*

$$N = d\,w^2\,(a + 2b), \qquad e \mid w\,(c - a - b), \qquad \frac{w(c - a - b)}{e} \equiv N \pmod 2.$$

Symmetrically, with $a = d'e'^2$, the base-β target forces $N = d'w^2(2a + b)$, $e' \mid w(c - a - b)$ and the same parity for $w(c - a - b)/e'$.

Proof. By the area identity $Nb = k^2(a+2b)$ and $\gcd(a+2b, b) = 1$ we get $b \mid k^2$. For a prime $p \mid d$ the valuation $v_p(b) = 2v_p(e) + 1 \leq 2v_p(k)$ forces $v_p(k) \geq v_p(e) + 1$, and for $p \nmid d$, $v_p(k) \geq v_p(e)$; hence $de \mid k$, say $k = dew$, and substituting back, $N = dw^2(a + 2b)$. By Corollary 5 and the identity $(c - a - b)(c + a - b) = b(2b + a - 2c)$, the invariant value is $M = k(c - a - b)/b$, so $b \mid k(c - a - b)$, i.e. $de^2 \mid dew(c - a - b)$, i.e. $e \mid w(c - a - b)$; and $M = w(c - a - b)/e \equiv N \pmod 2$ by Corollary 5. $\square$

Theorem 7 is the degenerate case: N prime forces $d = w = 1$, so $b = e^2$ and $e \mid (c - a - b)$, which Lemma 6 refutes. The residual divisibility is governed by the following classification.

Proposition 12 (Solvability of $e \mid (a + b - c)$)**.** *Let $b = de^2$, d squarefree, $\gcd(a, b) = 1$. Then $e \mid (a + b - c)$ holds for a primitive $120°$-triple if and only if there is an integer j with $d < j < 2d$ such that ej is even,*

$$a = \frac{e^2(2d - j)(2d + j)}{4(j - d)}$$

is a positive integer coprime to de^2; in that case $c = a + \frac{e^2 j}{2}$. In particular, for $d = 1$ the range $(1, 2)$ contains no integer, which re-proves Lemma 6; for $d = 2$, $j = 3$, $e = 2$ one obtains $(a, b, c) = (7, 8, 13)$.

Proof. Since $e \mid b$, the hypothesis gives $c \equiv a \pmod e$, and $0 < c - a < b$ writes $c = a + et$ with $1 \leq t \leq de - 1$. Substituting into $c^2 = a^2 + ab + b^2$ and dividing by e, $a(2t - de) = e(de - t)(de + t)$. As $e \mid a(2t - de)$ and $\gcd(a, e) = 1$, we get $e \mid 2t$, say $2t = ej$ with $1 \leq j \leq 2d - 1$; substituting $t = ej/2$ gives $4a(j - d) = e^2(2d - j)(2d + j)$, whose right side is positive, forcing $j > d$ and the displayed formula. Conversely, given such j, setting $t = ej/2$, a as displayed and $c = a + et$ reverses the computation: $c^2 = a^2 + ab + b^2$ and $a + b - c = e(de - t)$ is divisible by e. $\square$

Theorem 13 (Spectrum lattice)**.** *Let (a, b, c) be a primitive 120°-triple, $b = de^2$ with d squarefree, $r = c - a - b$, $g = \gcd(e, r)$, $e_1 = e/g$, and*

$$T \;=\; \frac{r}{g} + d\,e_1^2\,(a + 2b) \pmod 2.$$

Set $E = e_1$ if T is even and $E = 2e_1$ if T is odd. Then the counts admissible for the base-α isosceles target (Theorem 11 with the parity clause) are exactly

$$N = d\,(Eu)^2\,(a + 2b), \qquad u = 1, 2, 3, \ldots$$

Proof. By Theorem 11 the conditions on w are $e \mid wr$ and $wr/e \equiv N \pmod 2$. As $\gcd(e_1, r/g) = 1$, the divisibility is equivalent to $e_1 \mid w$; writing $w = e_1 u$, the count is $M = ur/g$, and using $u^2 \equiv u \pmod 2$,

$$M - N = \frac{ur}{g} - d\,e_1^2 u^2 (a + 2b) \equiv u\Big(\frac{r}{g} + d\,e_1^2(a + 2b)\Big) = uT \pmod 2,$$

so the parity clause holds for all u when T is even, and exactly for even u when T is odd. □

Remark 14. The refined spectrum coincides with Zhang's constructed family $N = m^2 b(a + 2b)$ exactly when $E = e$; this holds for $(7, 8, 13)$ and $(5, 16, 19)$ (both have $E = e$), but not in general: in the parametrization $a = m^2 - n^2$, $b = 2mn + n^2$, an odd prime p with $p^2 \mid n$ makes p "free" ($v_p(r) \geq v_p(e)$, since $r = -n(m - n)$), and the lattice is coarser than $e\mathbb{Z}$. The smallest such tile is $(11, 24, 31)$ ($d = 6$, $e = 2$, $E = 1$): the value $N = 354 = 6 \cdot 59$ is admissible but lies *outside* the Zhang family $1416m^2$; likewise $(19, 261, 271)$ admits $N = 15689 = 29 \cdot 541$. These beyond-family admissible values are the new sharpest tests of sufficiency: either they are tiled by constructions not in [7], or a necessary condition beyond the invariant kills them.

Remark 15. Admissibility is necessary, not sufficient. The smallest admissible value is $N = 33 = 3 \cdot 11$ on the tile $(5, 3, 7)$ (where $d = 3$, $e = 1$, so the divisibility is vacuous); it is nevertheless excluded by Beeson's boundary analysis, which rules out $N < 36$ [3], while Herdt's 2673-tile tiling is the member $w = 9$ of the same family $33w^2$. The parity clause bites hard: on the tile $(7, 8, 13)$ (where $d = 2$, $e = 2$, $c - a - b = -2$) the divisibility admits all $N = 2w^2 \cdot 23$, but the count $M = -w$ must have the parity of $N = 2w^2 \cdot 23$, which is even — forcing w even, say $w = 2m$; the refined spectrum is exactly $N = 184m^2 = m^2\,b(a + 2b)$ — *precisely* Zhang's constructed family [7] for this tile. In particular $N = 46$, previously the smallest value passing every known necessary condition on this branch, is *excluded* (Section 7 checks the remaining branches), in exact agreement with the prediction of Zhang's conjecture. Zhang's constructions realize $N = m^2 b(a+2b)$ for m in certain residue classes; the refined spectrum bounds the realized set from outside, and Theorem 13 determines it exactly.

Proposition 16 (The other branches)**.** *With the same normalizations: F_1 forces $N = dw^2(a+b)$ where $b = de^2$, d squarefree (from $b \mid k^2$ as above); F_2, F_3, F_4 force $N = N_0 k^2$ exactly, with $N_0 = (a + 2b)(2a + b)$, $3(a + b)(a + 2b)$, $(a + b)(2a + b)$; the tile-similar shape forces $N = n^2$; and an equilateral target of side S forces $ab \mid S^2$, $N = S^2/(ab)$ (so N and ab have the same squarefree kernel) together with $(c + a - b) \mid 3S$ and $(c + b - a) \mid 3S$.*

Proof. The scalene statements are the area identity (5) with the side-vectors of Section 3 (F_1 repeats the $b \mid k^2$ analysis). For the equilateral target the three sides, traversed counterclockwise, differ in direction by $\frac{2\pi}{3} = 2 \cdot \frac{\pi}{3}$, so f takes one sign on all three and $\Phi_f(\partial ABC) = \pm 3S$ for both invariants; Corollary 5 gives the two divisibilities, and the area identity gives $N\,ab = S^2$. □

The equilateral $2\pi/3$ spectrum. The equilateral divisibilities of Proposition 16 reduce to a clean finite criterion.

Proposition 17 (Equilateral admissibility)**.** *Let an equilateral triangle of side S be N-tiled by a $2\pi/3$ tile (a, b, c), and put $X = c + a - b$, $Y = c + b - a$. Then $XY = 3ab$, and $s := 3S/X$, $t := 3S/Y$ are positive integers with*

$$st = 3N, \qquad s \equiv t \equiv N \pmod 2, \qquad (t-s)^2 + 16N = q^2 \text{ for an integer } q;$$

conversely (s, t, q) determine the instance: (a, b, c) is proportional to $\big(q+(t-s),\ q-(t-s),\ 2(s+t)\big)$, and primitivity pins the scale, hence S.

Proof. $XY = c^2 - (a-b)^2 = 3ab$ by (1). By Proposition 16, $X \mid 3S$ and $Y \mid 3S$, so $s, t \in \mathbb{Z}_{\geq 1}$; indeed $s = M_\alpha$ and $t = M_\beta$ are the two invariant counts, so $s \equiv t \equiv N \pmod 2$ by Corollary 5, and $st = 9S^2/(XY) = 9\,Nab/(3ab) = 3N$. From $X - Y = 2(a-b)$ we get $a - b = \frac{3S}{2}\big(\frac{1}{s} - \frac{1}{t}\big) = \frac{S(t-s)}{2N}$, and $4ab = 4S^2/N$ gives $(a+b)^2 = (a-b)^2 + 4ab = \frac{S^2}{4N^2}\big[(t-s)^2 + 16N\big]$; as $a+b$ is an integer, $q := 2N(a+b)/S$ is a rational with integer square, hence an integer, and $(t-s)^2 + 16N = q^2$. Then $a - b$, $a + b$ and $c = (X+Y)/2 = \frac{S(s+t)}{2N}$ give the displayed proportions. □

The values 14 and 15. The machinery above, combined with the published branch theorems, decides every branch of the classification for $N = 14$ and $N = 15$ by finite computations (`verify_frontier.py`): the commensurable, tile-similar, right-angle and $\gamma = 2\alpha$ branches exclude both values by their known N-forms; the sporadic $2\pi/3$ shapes exclude both by the admissible spectra of Theorems 11 and Proposition 16; the $\pi/3$-tile equilateral case excludes both by Beeson's square criterion ($(9N - M^2)(N - M^2)$ is a perfect square for some $M < \sqrt{N}$ [5, Thm. 3], which fails for every admissible M); and of the five target shapes of the $3\alpha + 2\beta = \pi$ branch [2], four exclude both values through their published tiling equations (checked by exact rational-root enumeration), while the isosceles target with base angles $\alpha+\beta$ admits, for $N = 14$, the single candidate tile $(45, 56, 81)$ with $M = 2$ — which is destroyed by Beeson's boundary congruence $M \equiv -m \pmod{\gcd(a, c)}$ (it forces at least seven b-edges of length 56 on a base of length 140).

What survives everything is four fully explicit finite instances:

N	branch	tile	target
14	equilateral, $2\pi/3$	$(7, 8, 13)$	equilateral, $S = 28$
15	equilateral, $2\pi/3$	$(3, 5, 7)$	equilateral, $S = 15$
15	$3\alpha + 2\beta = \pi$, iso-$(\alpha+\beta)$, $M = 1$	$(56, 15, 64)$	$(120, 120, 105)$
15	$3\alpha + 2\beta = \pi$, iso-$(\alpha+\beta)$, $M = 3$	$(4, 15, 16)$	$(60, 60, 15)$

In each, the tile is *uniquely determined* (Proposition 17 for the equilateral rows; the tiling equation $N/M^2 = (1+s)/(1-s)$ of [2, Thm. 17] for the isosceles rows), so each is a single bounded existence question: can 14 (resp. 15) congruent copies of the stated tile fill the stated triangle?

Each instance was settled by an exhaustive, exact search (`code/engine/`). The engine is an advancing-front algorithm over $\mathbb{Q}(\sqrt{D})$ ($D = 3, 23, 7$; every geometric decision exact, no floating point) whose branching rule is provably complete: at the lexicographically least vertex v of the unfilled region — necessarily convex — every tile of any tiling whose closure contains v has a corner there (an edge through v would subtend angle π), so the clockwise-most of them has a corner at v with one side along the clockwise boundary ray; this leaves at most six placements (three corner types, two adjacent sides each, the two choices being mirror images). An exhausted search is therefore a proof of non-existence. Two prunes are used, each a theorem about completable states: the area of every region component is a positive integer multiple of the tile area; and a maximal straight boundary run with *convex* endpoints has length in $\mathbb{N}a+\mathbb{N}b+\mathbb{N}c$ (whole tile edges cannot pass a convex corner — runs with a reflex endpoint are not pruned, as an edge may continue past a reflex corner into the region). The engine passes positive controls — it finds the $N=4$ reptile subdivisions in both tile families and a hand-built non-edge-to-edge two-tile configuration with T-junctions, each found tiling re-verified by an independent checker — and, as a robustness check, all four verdicts below are unchanged when the run prune is disabled entirely (node counts without it in parentheses).

instance	tile	target	verdict	nodes
$N=14$, equilateral	$(7,8,13)$	$S=28$	exhausted, no tiling	133 (482)
$N=15$, equilateral	$(3,5,7)$	$S=15$	exhausted, no tiling	156 (604)
$N=15$, iso-$(\alpha+\beta)$	$(56,15,64)$	$(120,120,105)$	exhausted, no tiling	37 (526)
$N=15$, iso-$(\alpha+\beta)$	$(4,15,16)$	$(60,60,15)$	exhausted, no tiling	2 (10)

Theorem 18. *No triangle can be cut into* 14 *congruent triangles, and none into* 15.

Proof. By the branch sweep above, any 14- or 15-tiling would live in one of the four listed instances, each with its tile and target uniquely determined; the exhaustive searches refute all four. The dependencies are those of Theorem 1 together with [2, Thms. 3, 7, 11, 14, 17, 19], [5, Thm. 3], and the declared computer assistance (exact arithmetic, complete branching, independently validated). □

The values 21, 22, 30, and the instance 46. The same sweep, sharpened by the parity clause of Corollary 5, extends further (`verify_frontier.py`). Three remarks carry the new weight. First, the invariant applies to the *scalene* F_1 target: its sides $k(a+b), ka, kc$ have $\frac{\pi}{3}$-coefficients $0,2,3$ (the same turning computation as Lemma 4), so

$$M_\alpha = \frac{k(2a+b-c)}{c+a-b}, \qquad M_\beta = \frac{k(2a+b+c)}{c+b-a}$$

must be integers of the parity of N. For $N=21$ this destroys the unique structural candidate, the genuine tile $(5,16,19)$ with $k=4$: $M_\alpha = 28/8 \notin \mathbb{Z}$ — while Zhang's constructed F_1 family $N=m^2b(a+b)$ passes, so the invariant separates the candidate from the constructions. (For $N=30$ the candidate $(7,8,13)$, $k=4$ has $M_\alpha=3$, killed by parity.) Second, for $N=22$ and $N=30$ *every* branch dies arithmetically — the two isosceles-$(\alpha+\beta)$ candidates of 22 fall to the boundary congruence, those of 30 likewise — so no search is needed at all. Third, for $N=21$ exactly two instances survive, both isosceles-$(\alpha+\beta)$: $M=1$, tile $(110,21,121)$ on $(231,231,210)$, and $M=3$, tile $(10,21,25)$ on $(105,105,42)$; both live in $\mathbb{Q}(\sqrt{6})$, and the engine exhausts both:

instance	tile	target	verdict	nodes
N=21, iso-$(\alpha+\beta)$, M=1	$(110,21,121)$	$(231,231,210)$	exhausted	107 (4266)
N=21, iso-$(\alpha+\beta)$, M=3	$(10,21,25)$	$(105,105,42)$	exhausted	15 (102)

Finally, the parity clause settles the value that Remark 15 identified as the sharpest test of Zhang's conjecture: $N=46$ passes the divisibility spectrum on $(7,8,13)$ but not the parity, and the sweep excludes every other branch, so 46 is not realizable — exactly as Zhang's conjecture predicts.

The pipeline is now mechanical, and it continues: $N=38$ dies in every branch by arithmetic alone, while 33, 35 and 39 each reduce to two isosceles-$(\alpha+\beta)$ instances (the sporadic-$2\pi/3$ candidate $(5,3,7)$ at $N=33$ is excluded by Beeson's $N<36$ boundary analysis [3]); all six searches exhaust:

instance	tile	target	nodes
N=33, M=1	$(272, 33, 289)$	$(561, 561, 528)$	489
N=33, M=3	$(28, 33, 49)$	$(231, 231, 132)$	49
N=35, M=1	$(306, 35, 324)$	$(630, 630, 595)$	523
N=35, M=5	$(6, 35, 36)$	$(210, 210, 35)$	2
N=39, M=1	$(380, 39, 400)$	$(780, 780, 741)$	853
N=39, M=3	$(40, 39, 64)$	$(312, 312, 195)$	67

Theorem 19. *No triangle can be cut into* 21*,* 22*,* 30*,* 33*,* 35*,* 38*,* 39*, or* 46 *congruent triangles.*

Proof. The sweeps above; for 21, 33, 35 and 39, additionally the exhaustive searches listed. Dependencies as in Theorem 18. □

Decidability. The pattern of the last two subsections is not an accident. Every branch of the classification now reduces, for each fixed N, to a *finite* list of fully determined instances: the commensurable, similar-tile, right-angle and $\gamma = 2\alpha$ branches are decided by closed N-forms; in the sporadic $2\pi/3$ shapes the divisor structure of N (Theorems 11, 13, Proposition 16) leaves finitely many (tile, scale) pairs; the equilateral cases pin the tile from finitely many factorizations (Proposition 17) or coloring numbers ([5, Thm. 2], which determines the tile from (N, M)); and the five $3\alpha + 2\beta = \pi$ shapes have finitely many (M, s) candidates by their tiling equations [2]. Each surviving instance is a bounded existence question, and the corner-anchored search of `code/engine/` decides it in finitely many steps (the branching is provably complete and the depth is N). Hence:

Theorem 20 (Decidability)**.** *It is decidable, given N, whether some triangle can be cut into N congruent triangles (under the same cited inputs as Theorem 1).*

This extends the decidability that Beeson established for equilateral targets [5, Thm. 4] to the whole problem. What decidability does *not* give is a closed-form description of the set: that residual question concentrates in the equilateral branch, where Laczkovich has related the possible N to rational points on elliptic curves [11], and in the sufficiency half of Zhang's conjecture, for which the beyond-family values of Remark 14 (the smallest being $N = 354$) are now the sharpest tests.

State of the full problem. Assembling: the commensurable branch is completely determined (squares, sums of two squares, and $2, 3, 6$ times squares, all realized [4]); the similar-tile and right-angle branches are settled in [1, 12]; the primes are settled by Theorem 10. What remains open for the full determination of the tile-count set is: sufficiency in the sporadic $2\pi/3$ branches (Zhang's completeness conjecture [7]; the admissible spectra above are the outer bounds), the $3\alpha + 2\beta = \pi$ and $\gamma = 2\alpha$ branches beyond the partial results of [2, 3], and the equilateral targets for incommensurable tiles, where Laczkovich has connected the possible N to rational points on elliptic curves [11]. With 7 and 11 excluded by [4], the primes by Theorem 10, and $14, 15, 21, 22, 30, 33, 35, 38, 39, 46$ by Theorems 18 and 19, the membership of every $N \leq 41$ is now determined; the smallest undetermined values, to our knowledge, are $N = 42$ and $N = 44$.

Standing of the inputs. For the record, the external inputs to Theorem 1 and their publication status: Laczkovich's classification papers [9, 10] and the rep-tiling theorem of Snover–Waiveris–Williams [12] are refereed journal articles, and Beeson's isosceles paper [3] is accepted (to appear in *Integers*); Beeson's remaining branch papers [1, 2, 4, 5] and the Beeson–Zhang rationality theorem [6], which is load-bearing for the integrality of the tile, are arXiv preprints at the time of writing. Zhang's constructions [7] are cited for context only and carry no logical weight here. The parts original to this paper (Sections 3–5 and 7, together with the shape enumeration of Section 2) are self-contained given those inputs, with the arithmetic layer machine-checked (Section 8).

8. Verification

The finite and numerical claims are checked by the scripts in the repository (exact arithmetic). `verify_shapes.py` reproduces the eleven-shape enumeration of Section 2 and the values N_0 of Section 3. `verify_invariant.py` confirms Lemma 4 ($C_f(t) = \pm(c + a - b)$ over all orientations of T), Lemma 3 (both sides agree on explicit subdivision tilings, and (6) cancels at a non-edge-to-edge incidence), and Lemma 6 (no counterexample among primitive 120°-triples up to $c \approx 9 \times 10^6$). `verify_spectrum.py` checks Section 7: the two-positive-squares decomposition for all primes $p \not\equiv 3 \pmod 4$ below 10^5; the scale structure $b \mid k^2 \iff de \mid k$; the admissible spectrum of Theorem 11 (no prime is admissible; 33 and 46 are); and that every instance of $e \mid (a + b - c)$ with $a, b < 3000$ arises from the j-classification of Proposition 12, with none for $d = 1$. As a positive control, both necessary conditions of Theorem 7 hold on the genuine 2673-tile isosceles tiling of Herdt [3]: for its tile $(5, 3, 7)$ and scale $k = 27$, $N = k^2(a + 2b)/b = 2673$ and $M = k(2b + a - 2c)/(c + a - b) = -9$ are integers, as they must be on any true tiling.

The entire *arithmetic and combinatorial* layer of the proof is formalized and machine-checked in Lean 4 with Mathlib (`lean/Erdos634.lean`, twenty-three theorems); every theorem depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`:

> `shape_enumeration`: the eleven-triple completeness of Section 2 — realizable corner types with $\sum m_i = 0$, $\sum k_i = 3$ are exactly the eleven listed — as pure integer arithmetic;
> `k_not_dvd_sum_sub`, `M_not_int`: Lemma 6 ($(c - a - b)/\sqrt{b} \notin \mathbb{Z}$);
> the full arithmetic of Theorem 7 in one master statement: no prime N satisfies the 120°-relation, the area equation $Nb = k^2(a + 2b)$, and the integrality $(c + a - b) \mid k(2b + a - 2c)$ simultaneously (`iso_reduction_identity`, `prime_count_forces_scale`, `no_prime_isosceles_count`);
> `add_not_prime`, `not_prime_of_two_le`, `F1_count_not_prime`–`F4_count_not_prime`: the scalene counts of Proposition 2 are composite ($a + b$ is never prime for a primitive 120°-triple, and the F_2, F_3, F_4 counts factor);
> `prime_three_mod_four_excluded`: a prime $p \equiv 3 \pmod 4$ with $p > 3$ is neither a square, a sum of two squares, nor $2n^2$, $3n^2$ or $6n^2$ (the commensurable branch of Section 2), via Fermat's two-squares theorem;
> `prime_sum_two_pos_squares`: a prime $p \not\equiv 3 \pmod 4$ is a sum of two *positive* squares (the achievability half of Theorem 10);
> `iso_admissible`: Theorem 11 — given $b = de^2$ with d squarefree, the area equation and Φ-divisibility force $k = dew$, $N = dw^2(a + 2b)$ and $e \mid w(c - a - b)$;
> the finite arithmetic of the frontier sweeps, in eight theorems: the $\pi/3$-equilateral square criterion fails at 14 and 15; the first tiling equation has no admissible solution there; the two equilateral factor-pair uniqueness statements; the boundary congruences that kill the isosceles candidates at 14 and 22; the F_1 invariant kill at 21; and the parity kill of 46.

What is *not* formalized is exactly the geometry: the direction grid of Section 2 and Lemmas 3 and 4, which are proved in the text and checked numerically; Mathlib has no theory of planar dissections.

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and numerical verification, found the elementary proof of Lemma 6, and produced the Lean formalization, under the author's direction and review. The argument depends on the cited classification of Laczkovich and Beeson for the branches other than the $2\pi/3$ tile, and on the rationality theorem of Beeson and Zhang; the shape list coincides with Zhang's six sporadic types [7]. The present contribution is the invariant of Section 4, the consequent exclusion of *every* prime for the isosceles $2\pi/3$ branch (where Beeson had ruled out only $N < 36$ and left primality open), and the self-contained reduction of the scalene types. Its arithmetic layer is machine-checked (Section 8); the two geometric lemmas rest on written proofs supported by numerical checks, and the whole is offered for independent refereeing.

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